

Chapter: 18

MAGNETIC FIELDS

INTRODUCTION:

A magnetic field is a fundamental concept in physics that describes the region around a magnet or a current-carrying conductor where magnetic forces are exerted on other magnets, conductors, or charged particles.

Permanent magnets, such as iron magnets, have their own intrinsic magnetic fields due to the alignment of their magnetic domains. Magnetic fields have numerous practical applications, including:

- | | | |
|-------------------------------|-----------------------|---|
| 1. Electric motors | 2. Transformers | 3. MRI machines (Magnetic Resonance Imaging) |
| 4. Magnetic levitation trains | 5. Magnetic compasses | 6. Magnetic data storage (hard disk drives) |

MAGNETIC FIELDS:

MAGNETIC FIELD

Definition:

"An area around a permanent magnet or a current carrying conductor, where they generate a magnetic force is called Magnetic Field".

Explanation:

Unlike in electrostatics, where electric charges create an electric field and exert electric forces on other charges, the magnetic field lacks a direct counterpart due to the absence of magnetic monopoles.

Magnetic fields can be generated through two primary methods:

1. From permanent magnets.
2. From current carrying conductors, also known as electromagnets.

ELECTROMAGNETIC FIELD:

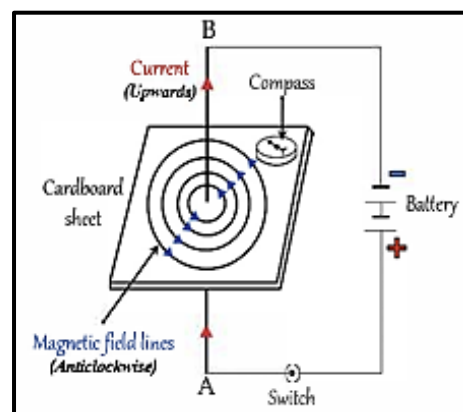
In 1819, Hans Christian Oersted made a significant discovery that revealed how a current carrying conductor generates a magnetic field.

Experiment:

The experiment shown in Figure demonstrates this phenomenon.

- Simple setup consists of a wire (AB) passing through a flat cardboard and connected to a battery via a switch.
- When the switch is closed, completing the circuit and enabling current flow through the wire, a circulating magnetic field is generated around it.
- When iron fillings are scattered on to the cardboard, they become magnetized and arrange themselves in a circular pattern around the current-carrying wire.
- Additionally, a magnetic compass placed near wire AB indicates the direction of the magnetic field.

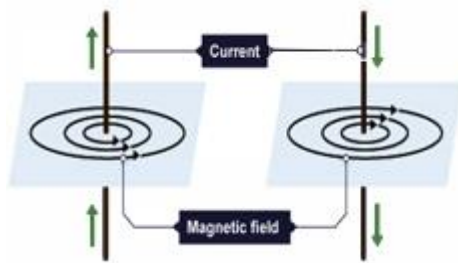
These observations illustrate that the magnetic field's direction produced by the current carrying wire follows the right-hand rule.



RIGHT HAND RULE:

According to this rule:

When a wire is grasped in the right hand, with the thumb pointing in the direction of the current, while the curl of fingers indicates the direction of the magnetic field line. If the direction of the current is reversed, the needles also reverse their orientation.



Conversely, if the current's direction is reversed, it generates a magnetic field in a clockwise direction. As shown in figure.

MAGNETIC FORCE ON A CURRENT CARRYING CONDUCTOR:

Electric current is the orderly flow of charges through a conductor. As electrons move, they create a magnetic field around them, in addition to the electric field they produce.

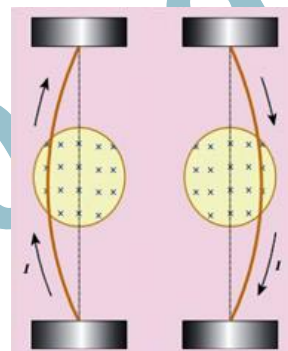
When it comes to a current-carrying conductor, the electric field generated by the moving electrons is counterbalanced by the electric field produced by stationary protons within the conductor.

Consequently, a magnetic field is produced around the current-carrying conductor.

When we place a current-carrying conductor with in a uniform external magnetic field, the interaction between the magnetic field produced by the conductor and the external magnetic field gives rise to external force, denoted as “F”, acting on the conductor, as illustrated in figure.

A vertical wire is securely anchored at both ends and passes through a gap in a magnet. The external magnet generates a uniform magnetic field directed into the page.

- When a current flows upward through the wire, the wire deflects to the left.
- Conversely, reversing the current causes the wire to deflect to the right.

**FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD:**

The force acting on a current-carrying conductor depends on several factors, such as, length, current, and the strength of the external magnetic field.

1. The force (F) is directly proportional to the length of the conductor (L) that lies within the magnetic field

$$F \propto L$$

2. The force is also directly proportional to the current (I) passing through the conductor.

$$F \propto I$$

3. Similarly, the force is directly proportional to the strength of the external magnetic field (B).

$$F \propto B$$

Combining these factors, we will obtain the following relationship:

$$F \propto BIL$$

Here, 'k' is the proportionality constant, and in the SI unit system, its value is equal to 1.

Therefore, we can express the force as: $F = BIL$ (1)

If the magnetic field is not perpendicular to the wire, Equation (1) can be modified to:

$$\vec{F} = I(\vec{L} \times \vec{B})$$

where 'X' denotes the vector cross product.

The magnitude of the force (F) is determined by:

$$|\vec{F}| = I|\vec{L} \times \vec{B}|$$

$$F = ILB \sin \theta$$

OR

$$F = BIL\sin\theta$$

Where, θ represents the angle between the direction of the conductor's length (L) and the external magnetic field (B).

1. MAXIMUM FORCE:

The deflecting force reaches its maximum, when the angle (θ) between the length of the conductor and the external magnetic field is $\theta = 90^\circ$ degrees, assuming other factors remain constant.

2. MINIMUM FORCE:

The deflecting force is minimized, when $\theta = 0^\circ$, signifying that the magnetic field and conductor are aligned.

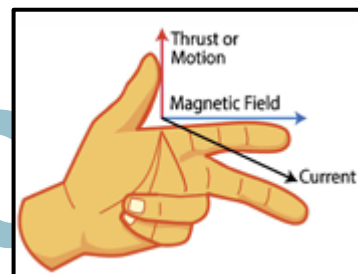
FLEMING'S LEFT-HAND RULE:

Since magnetic force is a vector quantity, its direction can be determined using Fleming's left-hand rule, which provides a straightforward way to establish the direction of the force in relation to the current and magnetic field.

Fleming's left-hand rule is used to determine the direction of force acting on a current carrying wire placed in a magnetic field, also to identify the direction of the forces in an electric motor.

To apply Fleming's left-hand rule, follow these steps:

Extends our fore finger, thumb, and Middle finger such that they are mutually perpendicular as shown in figure.



WORKED EXAMPLE:

18.11: A current carrying conductor of certain length placed at 90° to the magnetic field experiences a force F. What will be the force if the current is increased four times, length is halved, and magnetic field is tripled?

MAGNETIC FLUX:

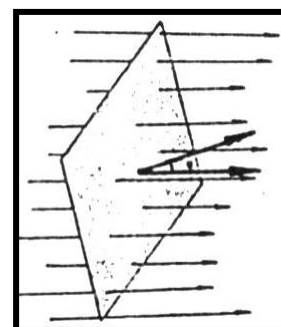
The total number of lines of induction passing through a surface placed in a magnetic field measures the magnetic flux through that surface.

Definition:

Magnetic flux is denoted by ϕ and it can be defined as:

"The number of magnetic lines of force passing through an area held perpendicular to it is called Magnetic flux". OR

"The dot product of magnetic field of induction "B" and vector area "A" through which magnetic lines of force are passing, is called magnetic flux" i.e.



$$\phi = \vec{B} \cdot \vec{A}$$

$$\phi = BA\cos\theta$$

FIGURE FROM LECTURE

Where " θ " is the angle between the direction of "B" and vector area of the surface.

Factors depending on Magnetic Flux:

The magnetic flux through a surface depends upon

1. The magnetic induction "B" of the field
2. The area of surface "A", and
3. The angle between magnetic induction and vector area.

Explanation:

It is calculated by taking the dot product between the magnetic field and area vector. The dot product ensures that magnetic lines must pass perpendicular to the given area.

Experimentally magnetic flux is usually measured with a Flux meter.

Magnetic flux through area A placed in uniform magnetic field is calculated by using formula as given below

$$\phi_B = \vec{B} \cdot \vec{A}$$

$$\text{Or } \phi_B = BA \cos \theta \dots (18.2)$$

Where B is uniform magnetic field, A is an area vector whose magnitude is equal to the surface element and directed perpendicular to the given surface element.

Unit of Magnetic Flux:

The SI unit of magnetic flux is Weber (Wb).

$$1\text{Wb} = \text{Nm} / \text{A}$$

It is named after renowned German Physicist and co-inventor of telegraph.

$\phi_B = B \cdot A$, represents the magnetic flux ϕ_B through a surface (A) in a magnetic field (B).

1. MAXIMUM MAGNETIC FLUX:

The magnetic flux will be maximum when the angle between magnetic field B and area vector A is zero, i.e. $\theta = 0^\circ$ and surface will be perpendicular to the magnetic lines. This case is shown in figure (a).

If $\vec{B}$ is perpendicular to the surface of element, then $\theta = 0$

Since

$$\phi = B A \cos \theta$$

$$= B A \cos 0$$

$$= B A \times 1$$

$$\phi = B A$$

It means that flux is maximum at $\theta = 0^\circ$.

2. MINIMUM MAGNETIC FLUX:

Similarly, when the angle between area vector and magnetic field is 90° then lines of force do not pass through the surface and magnetic flux will be minimum as shown in figure (b).

If $\vec{B}$ is parallel to the surface of area, then $\phi = 90^\circ$.

Since

$$\phi = B A \cos \theta$$

$$= B A \cos 90$$

$$= B A \times 0$$

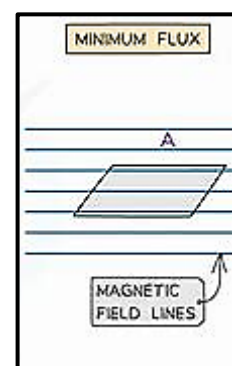
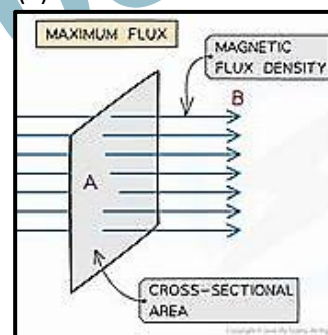
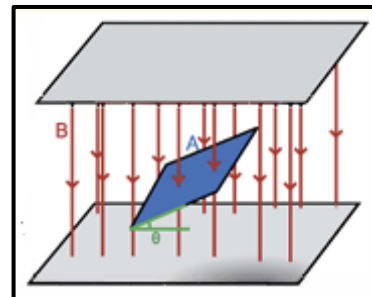
$$\phi = 0$$

It means that flux is zero at $\theta = 90^\circ$.

MAGNETIC FLUX DENSITY:**Definition:**

“The magnetic induction “B” is numerically equal to flux passing per unit area of the surface held perpendicular to “B”

As “B” is defined as “flux per unit area” so it is termed as flux density.



Explanation:

Since, we know that, if a surface of area “A” is perpendicular to the magnetic field, then $\theta = 0$ and magnetic flux is maximum/

$$\begin{aligned}\phi &= \vec{B} \cdot \vec{A} \\ \phi &= B A \cos 0 \\ \phi &= B A \times 1 \\ \phi &= B A\end{aligned}$$

Or
$$B = \frac{\phi}{A}$$

This equation provides a definition of $\vec{B}$ in terms of magnetic flux.

RELATION BETWEEN MAGNETIC FLUX AND MAGNETIC FORCE:

Magnetic flux density is described in relation to the magnetic force experienced by a current carrying conductor placed perpendicularly to a magnetic field.

For a uniform magnetic field, the flux density B is expressed by the equation:

$$B = \frac{F}{IL}$$

Hence magnetic flux density can be defined as:

“The magnetic flux density at a specific point in space is the force experienced per unit length by along straight conductor carrying unit current, placed perpendicular to the field at that particular location”.

Magnetic flux density is most pronounced near the pole of a bar magnet and gradually decreases with distance from it.

Unit of Magnetic Flux Density:

The SI unit of magnetic flux density is Tesla (T), named after the Serbian American inventor Nikola Tesla.

One Tesla: is defined as:

If a conductor having length 1m and carries a current of 1A placed perpendicularly to the magnetic field experiences a force of one Newton, then magnetic flux density will be 1 Tesla”.

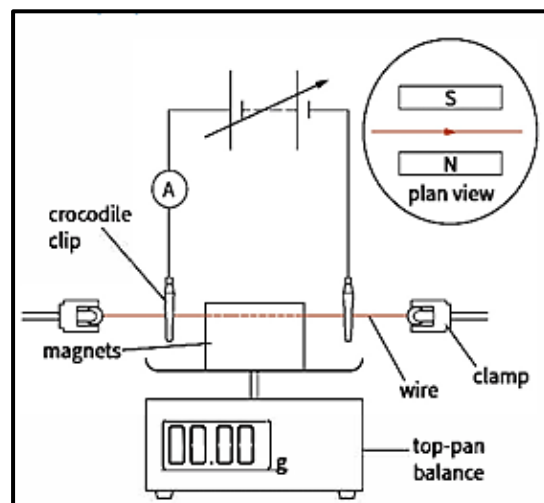
In essence, the Tesla provides a standardized measure of the strength of a magnetic field, with larger values indicating stronger magnetic fields and smaller values representing weaker fields.

MEASUREMENT OF MAGNETIC FLUX DENSITY BY CURRENT BALANCE:

A simple experimental setup designed for the measurement of flux density between two magnets, as shown in figure.

Experiment:

- The magnetic field within this magnet configuration is uniform.
- To measure the length (L) of the current carrying wire within this uniform magnetic field, a ruler can be used.
- When the wire carries no current, the magnet arrangement is positioned on top of balance, and the balance is calibrated to read zero.
- Subsequently, when an electric current (I) flows through the wire, the ammeter displays its magnitude.



- The wire experiences an upward force, and in accordance with Newton's third law of motion, an equal and opposite force acts upon the magnets.
- Consequently, the magnets are pushed downwards, causing the balance to indicate a reading.

The force (F) can be calculated as: $F = mg$

Where 'm' represents the mass indicated on the balance in kilograms and 'g' represents the acceleration due to gravity (9.81ms^{-2}).

With the knowledge of F, l, and L, one can determine the magnetic flux density (B) between the magnets using the following equation $B = F/IL$ (3)

WORKED EXAMPLE:

18.3 : Calculate the magnetic flux through a rectangular loop with dimensions 5 cm x 10 cm placed in a magnetic field of strength 0.2 T. The loop is oriented such that the magnetic field lines are perpendicular to the plane of the loop.

AMPERES LAW:

Introduction:

Ampere's Law is an important principle in electromagnetism. It deals with the behavior of electric currents and their associated magnetic fields.

It is named after Andre-Marie Ampere, a French physicist and mathematician who made significant contributions to the study of electromagnetism in the mid 19th century.

Statement:

"The sum of the products of the tangential component of magnetic field of induction and the length of an element on closed curve taken in the magnetic field is μ_0 times the current which passes through the area bounded by this curve".

Mathematical representation:

It can be represented mathematically as follows:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Where; μ_0 = Permeability of free space = $4\pi \times 10^{-7} \text{W/Amp.m}$

Explanation:

It is known that an electric current flowing through a wire creates a magnetic field in its vicinity.

When we envision a closed circular path with the wire positioned at its center, the Magnetic flux density (B) within this circular region changes, depending on both the current (I) and the distance (r) from the wire as shown in figure.

$$B \propto 2I \text{ (1)}$$

$$B \propto \frac{1}{r} \text{ (2)}$$

Combining (1) and (2), we get

$$B \propto \frac{2I}{r}$$

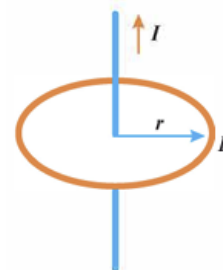
Or

$$B = \frac{\mu_0}{4\pi} \times \frac{2I}{r}$$

Where " $\frac{\mu_0}{4\pi}$ " is proportionality constant and its value is 10^{-7} and " μ_0 " is called the permeability of free space and its exact value is " $4\pi \times 10^{-7}$ ".

Hence

$$B = \frac{\mu_0 I}{2\pi r} \text{ (3)}$$



Now consider an Amperian loop in the form of a closed path around the current carrying wire as shown in figure.

For any path element ΔL , we can write $\vec{B} \cdot \vec{\Delta l} = \mu_0 I$

Since angle between B and L is zero so $B \cdot \Delta L = \mu_0 I$

Now adding up over whole closed path $\Sigma B \cdot \Delta L = \mu_0 I \dots \dots \dots (4)$

Equation (4) is the mathematical expression of Ampere law.

Hence Ampere's law can be defined as

"The integral of the magnetic field (B) around a closed loop is directly proportional to the total electric current (I) passing through the loop".

APPLICATION OF AMPERE'S LAW:

(1) SOLENOID:

Definition:

"A solenoid is a long, tightly wound coil of wire, and often used to generate a uniform magnetic field within the coil's interior".

Explanation:

It consists of many closely spaced turns of wire, and when an electric current flows through it, it creates a magnetic field that is similar to the field produced by a bar magnet.

To determine the magnitude of B of a solenoid, consider an Amperian loop "abcd" with lengths l_{ab} , l_{bc} , l_{cd} and l_{da} as shown in figure. Then from Ampere's law

$$\Sigma B \cdot \Delta l = \mu_0 I \dots \dots \dots (1)$$

The left-hand side of equation is

$$\Sigma B \cdot \Delta l = B \cdot l_{ab} + B \cdot l_{bc} + B \cdot l_{cd} + B \cdot l_{da}$$

As the element of lengths along bc and da are perpendicular to the magnetic field so

$$B \cdot l_{bc} = 0 \text{ \& } B \cdot l_{da} = 0$$

Moreover, the magnetic field intensity outside of solenoid ≈ 0 , so $B \cdot l_{cd} = 0$

Hence the magnetic field due to solenoid is given by

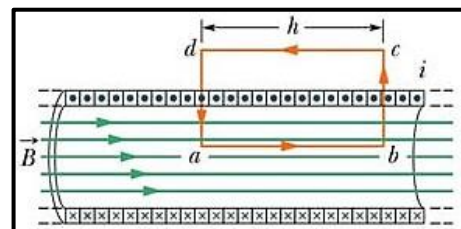
$$\Sigma B \cdot \Delta l = B \cdot l_{ab} = \mu_0 \cdot I$$

For N number of turns on the solenoid; $\Sigma B \cdot \Delta l = B \cdot l_{ab} = \mu_0 \cdot N I$

The number of turns per unit length is given by $n = \frac{N}{l_{ab}}$

$$B = \mu_0 \cdot n I$$

Hence it is shown that B is independent of the position within the solenoid which shows that the field is uniform within a solenoid.



WORKED EXAMPLE:

18.4: Suppose we have a solenoid with 500 turns per meter and a current of 2.5 amperes flowing through it. Calculate the magnetic flux density inside the solenoid.

(2) TOROID:

Definition:

"A toroid can be considered as a solenoid that is bent into the shape of a circle".

Explanation:

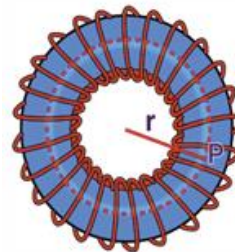
It also generates a magnetic field within its interior. The magnetic field lines inside a toroid are generally circular and concentrated within the coil.

Suppose a toroid consists of N closely packed turns of copper wire and carry a current! as shown in the figure.

Imagine a hollow circular ring wound with N number of turns of current-carrying wire. In this case, within the toroid, point P is our point of interest, and we will denote the magnetic field at this location as B .

Inside the toroid, we construct an Amperian loop forming a circle that passes through point P .

This arrangement gives rise to concentric circles within the toroid. Because of the symmetrical nature of the magnetic field, the magnitude of the field at all points along the circle is uniform, and the field direction is tangential to the circle. Therefore,



$$\begin{aligned}\Sigma B \cdot \Delta l &= \Sigma B \Delta L \cos 0 \\ &= B \Sigma \Delta L \\ &= B (2\pi r) \dots\dots\dots (1)\end{aligned}$$

Now, we can apply Ampere's law: $\Sigma B \cdot \Delta L = \mu_0 \cdot N I \dots\dots\dots (2)$

Therefore, we get $B (2\pi r) = \mu_0 \cdot N I$

$$B = \frac{\mu_0 N I}{2\pi r}$$

This is the expression for the magnetic field strength inside a toroid.

It is inversely proportional to the radius r and directly proportional to the number of turns per unit length N and the current " I " flow through the turns.

WORKED EXAMPLE:

18.5: Suppose we have a toroid with 1000 turns and a current of 3 amperes flowing through it. The toroid has a circular cross-section with a radius of 0.1 meters. Calculate the magnetic flux density inside the toroid.

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FORCE ON A CHARGED PARTICLE IN A UNIFORM MAGNETIC FIELD:

When a charged particle moves through a uniform magnetic field, it experiences a force known as the magnetic force. This phenomenon is described by the Lorentz force equation, which can be written as:

$$\vec{F} = q (\vec{v} \times \vec{B})$$

And the magnitude of force is $F = q v B \sin \theta$

IN THE EQUATION:

The magnitude of the charge (q):

The magnitude of the charge determines the strength of the magnetic force. A higher charge magnitude results in a stronger force.

Velocity of the Particle (v):

The magnetic force depends on both the magnitude and direction of the velocity vector. If the particle is stationary ($v = 0$), it would not experience a magnetic force, as there is no velocity to interact with the magnetic field.

Magnetic Field (B):

The magnetic field vector, denoted as B , represents the strength and direction of the magnetic field in which the charged particle is located.

A uniform magnetic field means that the strength and direction of the magnetic field are constant throughout the region. It is often represented by parallel magnetic field lines.

Cross Product ($\vec{v} \times \vec{B}$):

The key to understanding the magnetic force is the cross product ($\vec{v} \times \vec{B}$) in the Lorentz force equation.

The cross-product results in a force that acts perpendicular to both the velocity vector ($\vec{v}$) and the magnetic field vector ($\vec{B}$).

The direction of this force is determined by the right-hand rule, which states that:

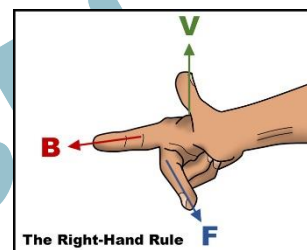
RIGHT HAND RULE:

If you point your right thumb in the direction of the particle's velocity ($\vec{v}$) and your fingers in the direction of the magnetic field ($\vec{B}$) then the force ($\vec{F}$) will be directed perpendicular to both, either into or out of the palm of your hand, depending on the charge of the particle.

When a charged particle moves through a uniform magnetic field, a magnetic force is exerted on it. This force is always perpendicular to both the particle's velocity and the direction of the magnetic field.

The charged particle will follow a curved path due to this force, known as a helical or circular trajectory, depending on the initial conditions and the specifics of the situation.

The magnitude of the force depends on the charge, the velocity, and the strength of the magnetic field.



CHARGE TO MASS RATIO OF ELECTRON:

Measuring the charge-to-mass ratio (e/m) of an electron is a classic experiment in the field of electromagnetism and particle physics. This experiment, known as the "e/m experiment", involves applying both a magnetic field and an electric field to a beam of electrons.

Here's a step-by-step method to measure the e/m of an electron:

APPARATUS AND MATERIALS:

1. Cathode Ray Tube (CRT):

This is a vacuum tube that produces a beam of electrons. It consists of an electron gun, focus in, deflection plates, and a fluorescent screen as shown in figure.

2. Magnetic Field Source:

We'll need a strong and uniform magnetic field source, such as a Helmholtz coil or a solenoid.

3. Voltage Source:

A variable voltage source to create an electric field.

4. Fluorescent Screen:

A screen coated with phosphorescent material to visualize the electron beam.

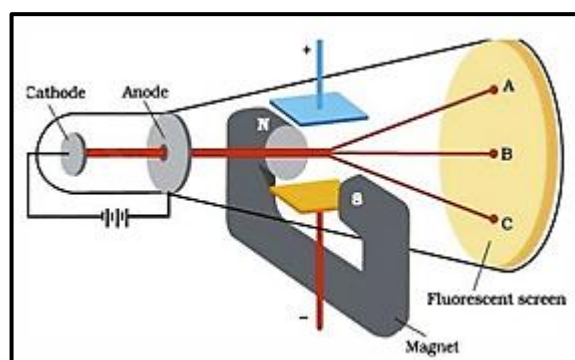
5. Rulers and Measurement Devices:

To measure the radius of the electron beam's circular path and the electric potential applied.

EXPERIMENTAL PROCEDURE:

1. Setup:

Set up the CRT in a vacuum chamber to ensure the electron beam doesn't interact with air molecules.



Position the Helmholtz coil (or solenoid) around the CRT, providing a uniform magnetic field parallel to the beam path.

Connect the voltage source to the focusing and deflection plates to create an electric field perpendicular to the magnetic field.

2. Calibration:

Calibrate the magnetic field by measuring its strength at the location of the electron beam. We can do this using a magnetic field strength meter.

3. Electron Beam Production:

Apply a high voltage across the cathode and anode of the CRT to generate a beam of electrons from the cathode (electron gun).

Use the focusing and deflection plates to control and direct the electron beam.

4. Observe the Electron Beam:

Turn off the magnetic field and electric field to observe the electron beam's straight-line path on the fluorescent screen.

5. Apply the Magnetic Field:

Turn on the magnetic field, which causes the electron beam to bend in a circular path due to the Lorentz force (the interaction between the magnetic field and the moving electrons).

6. Measure the Radius(r):

Measure the radius of the circular path formed by the electron beam on the screen. Ensure the screen is marked with a scale for accurate measurement.

7. Apply the Electric Field:

Apply a known electric potential (voltage) across the focusing and deflection plates perpendicular to the magnetic field. This will introduce an electric force on the electrons.

8. Adjust the Voltage:

Adjust the voltage until the electron beam returns to its original, undeflected straight-line path.

9. Measure the Electric Potential (V):

Measure the voltage applied across the plates.

CALCULATIONS:

An electron is released from cathode, and it gains speed while passing through the potential difference V , hence gaining kinetic energy of 1 eV. This electron kinetic energy is calculated by:

$$\frac{1}{2}mv^2 = Ve$$

The magnetic field produced by Helmholtz coils is perpendicular to this velocity, and produces a magnetic force which is transverse to both v and B , this provides the centripetal force makes an electron move along the circular trajectory; the radius of this trajectory r can be found from

$$\frac{mv^2}{r} = qvB$$

From this equation, we obtain the expression for the charge-to-mass ratio of the electron,

$$\frac{e}{m} = \frac{v}{Br}$$

Using the measured values for the radius (r), magnetic field strength (B), and electric potential (V), we can calculate the elm ratio of the electrons which was found to be $1.7588 \times 10^{11} \text{C/kg}$

MEASURING INSTRUMENTS:

GALVANOMETER:

A Galvano meter is an instrument used to detect and measure electric currents.

It is a highly sensitive electromagnetic apparatus capable of measuring even very small currents, such as those on the order of a few microamperes.

Principle:

It works based on the principle of electromagnetic induction.

When a coil carrying an electric current is positioned within an external magnetic field, it undergoes magnetic torque. The degree of deflection observed in the coil, caused by this magnetic torque, is directly proportional to the current's magnitude flowing through the coil.

CONSTRUCTION:

1. Coil:

The key component of a Galvano meter is a coil of wire (usually wound around a soft iron core) suspended within a magnetic field. The coil is mounted on a spindle so that it can rotate freely.

2. Magnet:

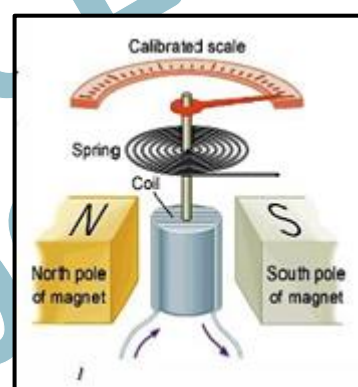
A permanent magnet or an electromagnet is placed around the coil. The magnetic field lines from the magnet pass through the coil.

3. Spring:

A delicate torsion spring is attached to the coil, providing a restoring torque when the coil is deflected.

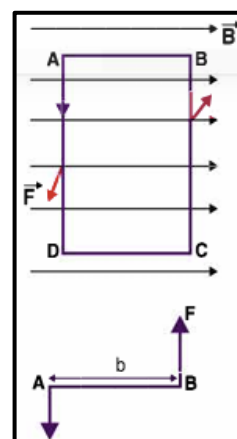
4. Pointer:

A thin pointer or needle is attached to the coil, allowing for the measurement of the deflection.



WORKING:

- When a small electric current flows through the coil, it generates a magnetic field around the coil according to Ampere's law. This magnetic field interacts with the external magnetic field (provided by the permanent magnet or electromagnet) to exert a torque on the coil.
- The torque causes the coil to rotate, and the amount of rotation is proportional to the current passing through it. This rotation is indicated by the deflection of the pointer on a calibrated scale.
- The coil continues to rotate until the restoring torque from the spring equals the torque due to the current-induced magnetic field. At this point, the pointer comes to rest, and its position on the scale indicates the magnitude of the current.
- Consider a rectangular coil consisting of N turns, each with a current (I) flowing through them and across-sectional area A .
- When this coil is situated within a uniform radial magnetic field B , it undergoes a torque as shown in figure.
- Let's examine a single turn ABCD of the rectangular coil, characterized by a length L and breadth b as shown in figure. This turn is suspended within a magnetic field with strength of B , arranged so that the coil's plane is parallel to the magnetic field lines.



- As sides AB and DC are aligned parallel to the magnetic field, they don't experience any discernible force due to the magnetic field.
- However, sides AD and BC, which are perpendicular to the field's direction, encounter an effective force denoted as F, given by the equation.

$$F = BIl$$

- By employing Fleming's left-hand rule, we can recognize that the forces acting on sides AD and BC are oriented in opposite directions to each other.
- When a pair of equal and opposite forces, denoted as F and collectively referred to as a couple, act on the coil, they generate a torque. This torque induces a deflection in the coil.
- The torque (τ) is calculated as the product of the force(F) and the perpendicular distance (b) between these forces:

$$\tau = F \times b$$

Substituting the known value of F, torque (τ) acting on a single-loop-ABCD of the coil.

$$\tau = BIL \times b$$

Where $L \times b$ represents the area (A) of the coil.

Consequently, the torque acting on a coil with n turns is given by:

$$\tau = NIAB$$

This magnetic torque causes the coil to rotate, leading to the twisting of the phosphor bronze strip. Simultaneously, the spring (S) attached to the coil exerts a counter-torque, known as the restoring torque ($k\theta$), resulting in a stable angular deflection. Under equilibrium conditions, we find:

$$k\theta = NIAB$$

Here, k is termed the torsional constant of the spring, representing the restoring couple per unit twist.

The deflection or twist (θ) is quantified as the reading displayed on a scale by a pointer connected to the suspension wire.

$$\theta = (NAB/k) I$$

$$\theta \propto I$$

Therefore, the quantity (NAB/k) is constant for a given galvanometer.

Therefore, it is evident that the deflection observed in the galvanometer is directly proportional to the current flowing through it.

CONVERSION OF GALVANOMETER TO AMMETER:

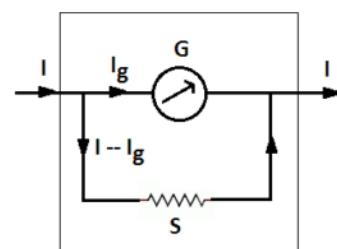
Conversion:

To convert a galvanometer into an ammeter, a low resistor known as a shunt is connected in parallel with the galvanometer.

Explanation:

The selection of an appropriate shunt is based on the desired current range of the ammeter. From the circuit as shown in figure.

- R_g represents the resistance of the galvanometer.
- G denotes the Galvanometer coil.
- I stand for the total current flowing through the circuit.
- I_g represents the total current passing through galvanometer, which corresponds to the full-scale reading.
- R_s represents the value of shunt resistance.



When the current I_g flows through the Galvano meter, the current passing through the shunt is determined by

$$I_s = I - I_g$$

The voltages across both the galvanometer and shunt are equal due to their parallel connection.

Hence, we can establish the following equation:

$$R_g I_g = (I - I_g) \times R_s$$

$$R_s = \frac{R_g I_g}{I - I_g}$$

WORKED EXAMPLE:

18.6: A moving coil galvanometer of resistance 100Ω is used as an ammeter using a Shunt of 0.1Ω . The maximum deflection current in the galvanometer is $100\mu\text{A}$. Find the current in the circuit so that the ammeter gives maximum deflection.

CONVERSION OF GALVANOMETER TO VOLTMETER:

Conversion:

A galvanometer is converted into a voltmeter by connecting it in series with high resistance.

Explanation:

A suitable high resistance is chosen depending on the range of the voltmeter, as shown in figure.

- R_g stands for the resistance of the galvanometer.
- R_x represents a high resistance component called series resistance.
- G denotes the galvanometer coil.
- I is the total current flowing through the circuit.
- I_g signifies the total current passing through the galvanometer, corresponding to a full-scale deflection.
- V represents the voltage drop across the series connection of the galvanometer and the high resistance.

When the current I_g flows through the series combination of the galvanometer and the high resistance R , the voltage drop across the branch ab can be expressed as:

$$V = V_g + V_x$$

The current series combination remains conserved, therefore. $I_g = I_x$

$$V = I_g R_g + I_g R_x$$

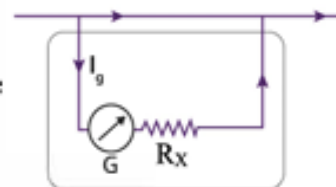
$$R_x = (V - I_g R_g) / I_g$$

$$R_x = \frac{V}{I_g} - R_g$$

This equation can be used to determine the value of R_x

WORKED EXAMPLE:

18.7: A galvanometer coil of 40Ω resistance shows full range deflection for a current of 4mA . How can this galvanometer be converted into a voltmeter range $0-12\text{V}$?



SUMMARY

- **Magnetic field:** The area around a permanent magnet or current-carrying conductor where a magnetic force is generated is called a magnetic field.
- **Magnetic force:** The force exerted on a wire carrying current i in a uniform magnetic field is given by the formula: $\mathbf{F} = \mathbf{BIL}\sin\theta$.
- **Measuring magnetic flux density:** A current balance can be used to measure magnetic flux density.
- **Ampere's law:** this law quantifies how a current creates a magnetic field around it, providing a mathematical relationship between current and magnetic field strength.
- **Solenoid:** a long, tightly wound coil of wire used to generate a uniform magnetic field within the coil's interior. The magnetic field inside a solenoid is given by: $B = \mu_0 nI$.
- **Toroid:** a solenoid curved to form a circular shape. The magnetic field strength inside a toroid is given by: $B = \frac{\mu_0 nI}{2\pi r}$. This formula shows that the magnetic field strength is $2\pi r$ inversely related to the radius r and directly related to the number of turns per unit length n and the current i passing through the turns.
- The shunt resistance for an ammeter is given by $R_s = \frac{R_g I_g}{I - I_g}$.
- In case of voltmeter the high resistance is given by $R_x = V - \frac{I_g R_g}{I_g}$.

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. If we reverse the direction of electric current, then the direction of magnetic field would be:

(a) Same
(b) magnetic levitation trains

(c) Tangent
(d) plant photosynthesis
2. The application of magnetic field is:

(a) Microwave oven
(b) reversed
(c) electrolysis
(d) normal
3. The equation $F = BIL$ can only be used, if the magnetic field, length of conductor, and electric current are:

(a) At right angles to each other
(b) in same direction

(c) Anti-parallel to each other
(d) anti-perpendicular to each other (2024)
4. The charge on a particle is doubled, and its velocity remains the same, how then the magnetic force on the particle will be:

(a) Doubled
(b) halved
(c) Is the same.
(d) Quadrupled (2025)
5. Strength of magnetic field of solenoid can be increased by:

(a) Increasing number of turns
(b) Decreasing number of turns

(c) Increasing the current through the solenoid
(d) Inserting a ferromagnetic core (e.g., iron) into the solenoid
6. The magnetic field inside along solenoid is:

(a) Equal to zero
(b) Uniform

(c) decreases as we go away from the center to surface
(d) increases as we go towards the surface
7. The force between two current carrying conductors arises due to

(a) Magnetic effect of current
(b) Polarization

- (c) Electromagnetic induction (d) Electrostatic interaction
8. To measure a higher voltage, what should you do with the voltmeter's internal resistance?
(a) Increase it (b) Decrease it
(c) Keep it the same (d) It doesn't affect the measurement
9. A proton moves perpendicular to a uniform magnetic field. What is the direction of the force experienced by the proton?
(a) Parallel to the magnetic field
(b) In the direction of the proton's velocity
(c) Perpendicular to the magnetic field and the proton's velocity
(d) Opposite to the direction of the proton's velocity
10. An electron moves parallel to a uniform magnetic field. What is the magnitude of the force experienced by the electron?
(a) Maximum, since the electron is moving in the same direction as the field
(b) Minimum, since the electron is moving perpendicular to the field
(c) Zero, since the electron is moving parallel to the field
(d) Depends on the speed of the electron

(MP:25)

SECTION- B: SHORT ANSWERED QUESTIONS

- Charge particles are fired in vacuum tube hit a fluorescence screen. Will it be possible to know whether they are positive or negative?
- What is a solenoid, and how does it differ from a simple coil of wire?
- Can a solenoid generate a magnetic field without any current flowing through it? Why or why not?
- Explain why a toroid is often preferred over a straight solenoid when designing certain types of electrical components.
- What role does the Ampere's circuital law play in understanding their magnetic field inside a toroid?
- What is a galvanometer, and what is its primary function in an electrical circuit?

SECTION- C (LONG ANSWERED QUESTIONS)

- Can a galvanometer measure both DC and AC currents? Explain any limitations it might have with AC measurements.
- Why should an ammeter ideally have low resistance compared to the circuits it is measuring?
- What is the potential risk of using a voltmeter with a high internal resistance in a circuit? How can this risk be mitigated?
- Describe the basic working principle of a voltmeter. How does it measure voltage across a component in circuits?

SECTION- D (NUMERICAL)

- An aluminum window has a width of 60 cm and length of 85 cm as shown in the figure. (a) When the window is closed the magnetic flux density is $1.8 \times 10^{-4} \text{ T}$ is normal to window. (b) Calculate the magnetic flux through the window. $(9.18 \times 10^{-5} \text{ Wb})$
- The poles of a horseshoe magnet measure 8cm x 3.2cm. The magnetic flux density between the magnet poles is 80mT. Outside of the magnet the magnetic flux density is zero. Calculate the magnetic flux density between the poles of a magnets. $(2.048 \times 10^{-4} \text{ Wb})$
- A wire 1.80 m long carries a current of 13.0 A and makes an angle of 35.0° with a uniform magnetic field of magnitude $B=1.50 \text{ T}$. Calculate the magnetic force on the wire. (19.046 N)

4. A solenoid has length $L = 1.23\text{m}$ and inner diameter $d = 3.55\text{ cm}$, and it carries a current $I = 5.57\text{ A}$. It consists of five close-packed layers, each with 850 turns along length L . what is B at its center.
(Ans: 24.2mT)
5. A moving coil galvanometer has resistance of 50Ω and it gave full scale deflection at 4mA current. A voltmeter is made using this galvanometer and a $5\Omega\text{ k}$ resistance. Calculate the maximum voltage that can be measured using this voltmeter. (Ans: 20.2V) (2025)
6. Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1A at distance of 1m from it compare it with Earth's magnetic field. (Ans: $2 \times 10^{-6}\text{ Weber/m}^2$; $39 \times 10^{-6}\text{ N}$)
7. Find the current in along straight wire that would produce a magnetic field twice the strength of the Earth's at a distance of 5.0cm from the wire. (Magnetic field of Earth = $5.0 \times 10^{-5}\text{ T}$)
8. What is flux density at a distance of 0.1m in air from along straight conductor carrying a current of 6.5 A. calculate the force per on a similar parallel conductor at a distance of 0.1m from the first and carrying a current of 3A. ($\mu_0 = 4\pi \times 10^{-7}\text{ T.m.A}^{-1}$) (Ans: $13 \times 10^{-6}\text{ Weber/m}^2$, $39 \times 10^{-6}\text{ N}$) (MP:25)

ASSIGNMENT: 18

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. The moving charge particle creates: (2024)
(a) Magnetic field only (b) Electric field only
(c) Both magnetic and electric field (d) Gravitational field only
- Magnetic Flux & Flux Density:**
2. tesla (T) is the unit of: (2021)
(a) potential difference (b) energy (c) torque (d) magnetic flux density
3. Tesla (T) is equal to: (2018)
(a) Newton x coulomb (b) Newton x ampere (c) $\frac{\text{Newton}}{\text{ampere} \times \text{meter}}$ (d) $\frac{\text{Newton} \times \text{meter}}{\text{ampere}}$
4. Which one is not a unit of magnetic flux density? (2009)
(a) $\text{N A}^{-1}\text{m}$ (b) Wb/m^2 (c) Tesla (d) Va^{-1}S
5. Which is not magnetic material? (2006)
(a) Iron (b) Nickel (c) Cobalt (d) Silver
6. The S.I. unit of magnetic flux is: (2006)
(a) Tesla (b) Weber (c) Gauss (d) Ohm
7. When the north pole of a bar magnet approaches the face of a closed coil the face becomes: (2003)
(a) South pole (b) First north pole and then south pole
(c) north pole (d) no effect is observed

Force on a current carrying conductor in the uniform Magnetic Field:

8. If the current passing through a wire held in a uniform field is doubled, the force acting on the wire will become: (2021)
(a) Half (b) Double (c) four times (d) Six times
9. If the conventional electric current is flowing from top to bottom in a wire, then the direction of the magnetic lines of induction will be: (2017)
(a) anti parallel to the wire (b) antilock wise
(c) parallel to the wire (d) clockwise
10. An electric current on passing through a conductor, produces around it: (2012, 03)

- (a) an electric field (b) a magnetic field
(c) both electric and magnetic field (d) first a magnetic field then an electric field
11. Two free parallel wires carrying current in the opposite direction: (2011)
(a) do not affect each other (b) attract each other
(c) repel each other (d) none of these
12. The force per unit length of a current carrying conductor in a uniform magnetic field is given by: (2005)
(a) $IBL \sin \theta$ (b) $IBL \cos \theta$ (c) $IB \cos \theta$ (d) $IB \sin \theta$
13. The maximum magnetic force will not act on a current carrying conductor in a magnetic field when it is placed. (2001)
(a) At 60° to the field. (b) Parallel to the field
(c) Perpendicular to the field (d) At angle of 45° to the field
14. The force experienced by a current-carrying conductor when it is placed in a magnetic field is: (2001)
(a) $\vec{F} = 1 (\vec{V} \times \vec{B})$ (b) $\vec{F} = 1 (\vec{L} \times \vec{B})$ (c) $\vec{F} = B (\vec{I} \times \vec{L})$ (d) None of the above

Force on a moving charged particle in the uniform Magnetic Field:

15. The equation $F = BIL$ can only be used, if the magnetic field, length of conductor, and electric current are:
(a) At right angles to each other (b) in same direction
(c) Anti-parallel to each other (d) anti-perpendicular to each other (2024)
16. When a charged particle enters in a magnetic field $\vec{B}$ with velocity $\vec{v}$ making angle θ , the trajectory of particle will be: (2022)
(a) circular (b) elliptical (c) helix (d) straight line
17. The path of neutron, moving perpendicularly through a uniform magnetic field is: (2019)
(a) A straight line (b) Circular (c) Oval (d) Sinusoidal
18. When an electron moves in a magnetic field (B) with a velocity (v), the magnetic force acting on it is perpendicular to: (2015)
(a) $\vec{V}$ but not on $\vec{B}$ (b) $\vec{B}$ but not on $\vec{V}$ (c) neither $\vec{V}$ nor $\vec{B}$ (d) Both $\vec{V}$ and $\vec{B}$
19. The path of a neutron, moving perpendicularly through a magnetic field, is: (2014)
(a) Straight line (b) Circular (c) Oval (d) Sinusoidal
20. A charged particle moving in the magnetic field B experiences a resultant force: (2008)
(a) proportional to the kinetic energy (b) in the direction of the field
(c) in the direction perpendicular to its motion and field (d) none of these
21. The path of a neutron moving normal to a magnetic field is: (2006, 05)
(a) A straight path (b) Circular path (c) An oval path (d) A sinusoidal path
22. If an electron and a proton enter into a magnetic field perpendicularly with the same momentum:
(a) The electron will be deflected more. (b) The proton will be deflected more
(c) both particles will be deflected equally (d) They will not be deflected at all (2002E)

Galvanometer:

23. The sensitivity of a galvanometer can be increased by: (2021)
(a) Increasing magnetic field (b) Decreasing area of coil
(c) Decreasing of number of turns in the coil (d) Making suspension wire thick
24. Concave magnetic poles, with a fixed soft-iron cylinder in a moving coil galvanometer, make the magnetic field: (2019)
(a) Weak (b) Radial (c) Zero (d) Infinite
25. In a moving coil galvanometer pole pieces of U-shaped magnet are made concave: (2017)

- (a) to make the field radial (b) to weaken the field
 (c) to increase the sensitivity of galvanometer (d) to decrease the sensitivity of galvanometer
26. The sensitivity of a galvanometer can be increased by increasing: (2008)
 (a) magnetic field (b) area of coil (c) number of turns (d) all of them
27. The unit of least count on a galvanometer scale represents (2006)
 (a) Division (b) Ohm (c) Volt (d) Henry
28. $I = \frac{C}{BNA} \theta$. Hence to increase the sensitivity of a galvanometer, we must decrease the value of: (2006)
 (a) θ (b) N (c) B (d) C

Voltmeter:

29. A voltmeter is an ideal one if its internal resistance is: (2022)
 (a) very large (b) large (c) small (d) infinite
30. To convert galvanometer into voltmeter: (2021)
 (a) A high value resistance is connected in series (b) A high value resistance if connected in parallel
 (c) A low value resistance if connected in series (d) A low value resistance is connected in parallel
31. This is a high resistance instrument: (2014)
 (a) Voltmeter (b) Ammeter (c) Galvanometer (d) Motor

Ammeter:

32. Ammeter is used to measure: (2021)
 (a) current (b) potential difference (c) resistance (d) capacitance
33. A moving coil galvanometer is converted into an ammeter by connecting to it: (2007, 02, 01)
 (a) Low resistance in series (b) High resistance in series
 (c) High resistance in parallel (d) Low resistance in parallel

SECTION- B: SHORT QUESTIONS**Magnetic Flux & Flux Density:**

1. Define magnetic flux. Under what condition will the magnetic flux be maximum? (2025)
2. Define Magnetic Field. Derive the relation for the magnitude of Magnetic Field of induction B. Give its units. (2004)
3. Differentiate between magnetic flux and magnetic flux density? (2001)
4. What do you mean by magnetic flux and flux density? Give their units. How are they related to each other?

Force on a current carrying conductor in the uniform Magnetic Field:

5. Derive an expression for the force experienced by a current-carrying conductor in a uniform magnetic field. (2023, 17, 15, 13, 11)

Force on a moving charged particle in the uniform Magnetic Field:

6. A charged particle is projected perpendicularly in a uniform magnetic field. Write down the expression for the force acting on it. (1996, 95, 93)

Ammeter:

7. How can a galvanometer be converted into an ammeter? Derive the relevant expression. (2018, 14)

Voltmeter:

8. How can a galvanometer be converted into a voltmeter? Derive the relevant mathematical expression. (2019, 16)

NUMERICAL PROBLEMS**Motion of Charged particles in the uniform magnetic field:**

1. A proton of charge 1.6×10^{-19} C and mass 1.67×10^{-27} kg is accelerated by a potential difference of 6×10^5 volts. Then it enters perpendicularly into a magnetic field of intensity 0.5 Tesla. Find the radius of the circular path of the proton. (Ans: 0.26 m) (2016)
2. A proton accelerated through 1000 volts is projected normally to a 0.25 tesla magnetic field. Calculate: (i) The kinetic energy of the proton on entering the magnetic field. (ii) The radius of the circular path of the proton ($m_p = 1.67 \times 10^{-27}$ kg). (Ans: 1.6×10^{-16} J, 0.01824 m) (2005)
3. How fast must a proton of mass 1.67×10^{-27} kg be moving if it is to follow a circular path of radius 2.0 cm in a magnetic field of 0.7 Tesla? ($m_p = 1.67 \times 10^{-27}$ kg) (Ans: 1.34×10^6 m/sec) (2004)
4. An electron is moving along a circle of radius 1.8×10^{-6} m. Calculate the speed of the electron on entering perpendicularly in a uniform magnetic field of 5.00 Tesla. Given $e = 1.6 \times 10^{-19}$ C & $m = 9.11 \times 10^{-31}$ kg. (Ans: 1.58×10^6 m/sec) (2003E, 1995)
5. An electron is accelerated by the potential difference of 1000 volts. It then enters into a uniform magnetic field of induction $B = 2.5$ weber/m² at an angle of 45° with the direction of the field; find the radius of the path of electron. ($q = 1.6 \times 10^{-19}$ C, $m = 9.1 \times 10^{-31}$ Kg) (Ans: 30.15×10^{-6} m) (2001)
6. A 10 eV electron is moving in a circular orbit in a uniform magnetic field of strength 10^{-4} web/m². Calculate the radius of the circular path. ($m = 9.11 \times 10^{-31}$ kg and $q = 1.6 \times 10^{-19}$ C). (Ans: 0.107 m, 1.875×10^6 m/sec) (99, 93)
7. Alpha particles are accelerated from rest at a P.D of 1 KV. They then enter a magnetic field of 0.2 T perpendicular to their direction. Calculate the radius. ($m = 6.68 \times 10^{-27}$ kg; $q = 2e$; $e = 1.6 \times 10^{-19}$ C) (Ans: 0.322m) (1997)

Ampere's Law:

8. Suppose we have a toroid with 1000 turns and a current of 3 amperes flowing through it. The toroid has a circular cross-section with a radius of 0.1 meters. Calculate the magnetic flux density inside the toroid. (2024)
9. What amount of current is required to produce a magnetic field of 37.2 Tesla at a distance of 2cm from a long straight current carrying wire? ($\mu_0 = 4\pi \times 10^{-7}$ Weber/Am) (2022)
10. A current of 6.25 amperes is maintained in a long straight conductor by a source. Calculate the force per meter on similar parallel conductor in air at a distance of 0.5 m from the first and carry a current of 2 amperes. ($\mu_0 = 4\pi \times 10^{-7} \frac{T.m}{A}$) (2018)
11. Find the current required to produce a field of induction $B = 2.152 \times 10^{-3}$ tesla in a 50 cm long solenoid having 4000 turns of wire ($\mu_0 = 4\pi \times 10^{-7}$, wb/A – m) (Ans: 0.5 A) (2012, 10)
12. A long solenoid is wound with 35 turns in 10 cm and carries a current of 10A. Find the magnetic field in it: ($\mu_0 = 4\pi \times 10^{-7}$ W/Am) (Ans: 4.39×10^{-3} T) (2006)
13. A solenoid 20 cm long has three layers of windings of 300 turns each. If a current of 3 amperes is passed through it, find the value of the magnetic field of induction B. (Ans: 16.96×10^{-3} T) (1998)
14. The inner & the outer diameters of the Toroid are 22 cm and 26 cm. If a current of 5.0 amp is passed which produces 0.025 tesla flux density inside core, find the approximate length of the wire wound on the toroid. ($\mu_0 = 4\pi \times 10^{-7}$) (Ans: 188.52 m) (2011)

Ammeter:

15. A galvanometer, whose resistance is 60 ohms, deflects full scale for a potential difference of 100 millivolts across its terminals. What shunt resistance must be connected to convert it into an ammeter of 5 ampere range? (2015)

16. A galvanometer of resistance 50 ohm gives full scale deflection with a current of 10 mA. A shunt of 0.05 ohm is connected in parallel to convert it to an ammeter. Find the range of the ammeter. (Ans: 10.01 A) (2012)
17. A galvanometer deflects full-scale with a voltage of 2.5 volts and has a total resistance of 0.5 ohm. What small resistance must be connected to measure 20A full-scale? (Ans: 0.167 ohm) (2006)
18. A maximum 50 mA current can be allowed to flow through a 19.8 ohm coil of a galvanometer. The galvanometer is to be used to measure 5 A maximum current. Calculate the length of a copper wire to be used as a shunt. The diameter of the wire is 4mm; the specific resistance (ρ) of the copper is 1.6×10^{-8} Ohm-m. (Ans: 157.1 m) (2005)

Voltmeter:

19. A 400 volts galvanometer has a resistance of 40,000 Ω . What additional series resistance must be connected to it to increase its range to 750 volts? (2022, 17)
20. A galvanometer, having resistance 50 Ω , deflects full scale for a potential difference of 100 mV across the terminals. What resistance should be connected to increasing its range to 50 Volts? (Ans: 24950 Ω) (2014, 10, 00, 97)
21. A galvanometer measuring up to 200 volts has a total resistance of 20,000 ohms. What additional series resistance must be connected to it to increase its range to 600 volts. (2013)
22. A moving coil galvanometer has a resistance of 25 Ω and it gives full scale deflection for a potential difference of 50 mV. If the galvanometer is to be converted into a voltmeter of reading up to 50 volts, what would be the resistance of the series resistor? (Ans: 24975 Ω) (1997)

SECTION- C: DETAILED QUESTIONS:

Ampere's Law:

9. State ampere's law. Derive an expression for the magnetic field of induction 'B' due to current passing through a long solenoid OR Toroid. (MP:25) (2025)
10. State Ampere's law. Apply this law to find the magnetic field of induction inside a solenoid. (2022, 19, 15)
11. State Ampere's law. Use it to derive the relation for the magnetic field of induction B at any point inside a current-carrying toroid. (2017)

Galvanometer:

12. Describe the construction and working of moving coil galvanometer. Also, show that the deflection produced in the coil is proportional to the current passing through it. (2022, 19, 15)
13. Explain the functions of the various parts of a moving coil galvanometer and prove that the current flowing is directly proportional to the angle of twist of the suspension. (2017)

Ammeter:

14. Why should an ammeter ideally have a very low resistance compared to the circuit it is measuring? (2024)